Prajwal Amte

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GitHub | LinkedIn | Portfolio

Education

The National Institute of Engineering, Mysuru – Information Science B.E. Dec 2022 – June 2025

Experience

Fidelity Investments | Associate Software Engineer Bengaluru | Aug 2025 – Present

- Building enterprise risk, governance, and compliance platforms using Spring Boot, Angular, PostgreSQL, Docker, and cloud-native engineering practices.
- Developing AI-assisted risk control systems using LLMs and agentic workflows to automate evidence analysis, control assessment, and compliance monitoring.
- Engineering backend services and deployment workflows for scalable enterprise applications, with focus on system reliability, integration, testing, and automation.

Fidelity Investments | Intern (Leap & Summer) Bengaluru | June 2024 – June 2025

- Built and tested full-stack applications using Spring Boot, Angular, MySQL, and Azure; developed serverless automation with AWS Lambda and maintained test coverage using JUnit and Mockito.

Skills

Languages:	C++, Python, Java, Rust, JavaScript, Bash
AI / ML:	Applied ML, Feature Engineering, Model Evaluation, PyTorch, LLM Fine-tuning (QLoRA/LoRA), RAG, Agentic AI, LangChain, HuggingFace
Frameworks:	Spring Boot, FastAPI, TensorFlow, ONNX Runtime, React, AngularJS, CMake
Infrastructure:	Docker, Kubernetes, Jenkins, CI/CD, Git, AWS (S3, Lambda), Azure, PostgreSQL, Redis, Linux

Projects / Open-Source

Minerva – Modern Chess Engine | GitHub *C++20, Python, PyTorch, CMake*

- Built a UCI-compliant C++20 chess engine with bitboard move generation, alpha-beta search, transposition tables, iterative deepening, pruning, and NNUE evaluation, reaching approximately 980K nodes/sec.
- Developed an end-to-end Python ML pipeline for data extraction, HalfKP feature engineering, PyTorch training, quantization, and C++/Python model verification with 163K+ passing assertions.

Calibrate – GPU Training Efficiency Analyzer | GitHub *Rust, NVML, CUDA, ONNX, TensorRT*

- Built a Rust CLI that attaches to live training jobs and computes real-time Model FLOP Utilization (MFU) across NVIDIA and Apple Silicon workloads, identifying system bottlenecks and optimization opportunities.
- Benchmarks p50/p99 inference latency across ONNX, TensorRT, llama.cpp, Candle, and TorchScript runtimes and estimates fine-tuning costs using live GPU pricing data.

AI Engineering Patterns | Website *LLM Systems, RAG, Agentic AI*

- Authored a structured pattern library for production AI systems covering RAG, agentic workflows, evaluation, observability, deployment strategies, and architectural trade-offs.
- Created reusable implementation patterns and code examples establishing a shared engineering vocabulary for building reliable, scalable, and cost-efficient AI applications.

Certifications

- Introducing Generative AI with AWS – **Udacity**